LORAZEPAM

Status epilepticus first-line IV

Store injection in fridge

CHEMICAL NAME (RS)-7-chloro-5-(2-chlorophenyl)-3-hydroxy-1,3-dihydro-2H-1,4-benzodiazepin-2-one

CLASS Intermediate-acting benzodiazepine

TRADE NAMES Ativan · Temesta · Lorivan

THERAPEUTIC USES / INDICATIONS

- Status epilepticus (first-line IV benzodiazepine).
- Acute anxiety; acute agitation (with antipsychotic).
- Pre-medication; alcohol withdrawal (in liver disease; no active metabolites).
- Chemotherapy-induced anticipatory nausea.

ADMINISTRATION

- **IV** slowly (dilute 1:1 with saline) into large vein over 2 min.
- **IM** (well absorbed); **PO**, **sublingual**.
- Injection stored at 2–8 °C, protected from light.

DOSAGES

- **Status epilepticus (adult):** 4 mg IV, repeat once after 10 min.
- **Children:** 0.1 mg/kg IV (max 4 mg).
- **Anxiety:** 1–4 mg/day in divided doses (short-term).
- **Agitation:** 0.5–2 mg IM/PO.
- **Elderly:** 0.5–1 mg.

PHARMACOKINETICS

- **Onset:** IV 2–3 min; IM 15–30 min; PO 30–60 min. **Duration:** anticonvulsant 6–12 h.
- Glucuronidated (no CYP; no active metabolites) — safer in liver disease, elderly.
- Excreted in urine. **t½:** 10–20 h.

PHARMACODYNAMICS

- **GABA-A** positive allosteric modulator → anticonvulsant, anxiolytic, sedative, amnesic.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Sedation, drowsiness.
- Dizziness, ataxia.
- Amnesia.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression, hypotension.
- Dependence and withdrawal.
- Paradoxical reactions; falls.

CONTRAINDICATIONS & CAUTIONS

- Respiratory depression; sleep apnoea.
- Myasthenia gravis; severe hepatic failure (encephalopathy).
- **Caution:** elderly, COPD, substance misuse.

KEY INTERACTIONS

- **Opioids, alcohol:** additive respiratory depression.
- **Valproate, probenecid:** ↑ levels.
- **Clozapine:** respiratory/cardiac arrest (rare).

NURSING MANAGEMENT

- Monitor RR, SpO_2, BP, GCS.
- Have flumazenil and airway equipment ready.
- Check fridge storage; dilute before IV.
- Fall precautions.

PATIENT EDUCATION

- Causes drowsiness and memory gaps; no driving or alcohol.
- Short-term use; do not stop suddenly after regular use.

KEY POINTS

- IV benzo of choice for status epilepticus (4 mg).
- Fridge storage.
- No active metabolites — safer in liver disease.
- Respiratory depression; dependence.